Avel Plugins — User Guide

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Avel is a free collection of eight mixing and mastering plugins with one shared look and one way of working. This guide explains what each plugin does and how to use it.

The plugins at a glance

Plugin	Type	Best for
Avel Tape	Tape echo	Vintage repeats, dub throws, spring reverb
Avel Time	Stereo delay	Clean to creative delays with filters, pitch and modulation in the loop
Avel Verb	Algorithmic reverb	Rooms, halls, plates and ambiences
Avel EQ	Parametric equalizer	Surgical and musical EQ, dynamic EQ, matching a reference
Avel Dynamic	Compressor	Vocals, drums, buses, sidechain ducking
Avel Ess	De-esser	Taming harsh "s" sounds on vocals and bright sources
Avel Ember	Multiband saturator	Warmth, drive and distortion per frequency band
Avel Apex	Mastering limiter	Final loudness and true-peak safety on the master

Installation

All plugins come in one installer. macOS gets VST3 and Audio Unit (Apple Silicon and Intel). Windows gets VST3 (64-bit).

macOS — open `Avel-Bundle-macOS.pkg`, then Continue → Install, and enter your password. Plugins go to `/Library/Audio/Plug-Ins`. If macOS says it cannot verify the developer:

- macOS 15 and newer: open the file once, close the warning, then go to **System Settings → Privacy & Security** and click **Open Anyway**.
- macOS 14 and older: right-click the file → **Open** → Open.

Windows — run `Avel-Bundle-Windows-Setup.exe`, then Next → Install. Plugins go to `C:\Program Files\Common Files\VST3`. If you see "Windows protected your PC", click **More info → Run anyway**.

After installing, restart your DAW and rescan plugins. All plugins appear under the manufacturer name **Avel**.

Controls shared by every plugin

Action	How
Turn a dial	Drag up or down, or use the mouse wheel
Fine adjustment	Hold Shift while dragging
Reset to default	Double-click the dial or point
Type an exact value	Click the value under the dial and type, e.g. `250 ms`, `4.5k`, `-12`
Compare two settings	**A / B** in the top bar; the copy button copies the current slot to the other one
Undo / redo	Arrows in the top bar, or Cmd+Z / Shift+Cmd+Z (Ctrl on Windows)
Presets	Arrows step through presets; click the preset name for the full list and **Save Preset**

User presets are saved inside `Documents/Avel` and can be copied to another computer.

Avel Tape — tape echo

A vintage tape echo in the spirit of classic Space Echo units: a tape loop with one record head and three playback heads, wow and flutter, tape saturation and a spring reverb.



Top bar

Preset field with arrows and a full list. **A / B** with the chain button to copy one slot to the other. The gear sets window size (60–150 %; you can also drag the corner). The ⋮ menu saves a preset, resets to Init, or opens the preset folder.

Transport

- **HEAD PATTERN** chooses which heads play: I = the delay time, II = 2×, III = 3×, plus I+II, II+III and ALL.
- **PING PONG** sends repeats alternately left and right (left repeats are drawn above, right repeats below).
- Lamps: **TAPE** = signal reaches the tape, **HEADS** = repeats are playing, **MOTOR** = the plugin is running.
- The tape window shows the current repeats on a ruler in bars. Reel speed follows the delay time.

Dials

Dial	What it does
Delay Time	Time of head I, 20 ms to 2 s. The note button picks a note value; **SYNC** locks to the project tempo (up to 1/1)
Feedback	Number of repeats. Above 100 % the loop self-oscillates (the lamp blinks); tape saturation keeps it in check
Mix	Dry/echo balance. The **DRY / WET** switch removes the dry signal for sends and aux tracks
Wow / Flutter	Slow drift and fast wobble of tape speed
Saturation	Drive at the record head, up to +18 dB; every repeat is saturated again
Tone	Colour of the repeats: DARK, WARM, NEUTRAL, BRIGHT, AIRY
Spring Reverb	Spring reverb on the input and the repeats

Bottom strip

- **Tape Mode** — CLEAN, VINTAGE, or WORN (dull, uneven, with dropouts).
- **Tape Type** — NORMAL, HIGH (brighter, more headroom), LOW (darker, saturates earlier).
- **Noise** — tape hiss from OFF to −30 dB. Hiss is recorded to the tape and repeats with the echo.
- **Age** — tape wear: dull top end, dropouts, uneven speed.
- **Flutter Mod** — turns all speed modulation on or off.

User presets go to `Documents/Avel/Avel Tape/Presets` as `.xml` files you can share.

Avel Time — stereo delay

A versatile stereo delay with two independent channels. Filters, saturation, pitch and diffusion sit inside the feedback loop, so every repeat is reshaped again. Switch between three pages with the segment at the top centre: **Delay**, **Filters** and **Modulation**.



Delay page

Repeats of the left channel are drawn above the line, the right channel below. Bar height shows each repeat's level. Drag the first repeat of either side to set its time; hold Shift to snap to whole milliseconds.

- **Sync** — time in note values from 1/64 to 4/1, including dotted (D) and triplet (T). **Link L/R** makes the right channel follow the left. **Tap** sets the time from at least two taps.
- **Routing** — Stereo, Ping-Pong (repeats bounce L → R → L) or Mono.
- **Time Change** — how the echo reacts when you change the time: **Tape** glides the pitch like tape, **Fade** crossfades, **Jump** switches quickly.

Dial	What it does
Time L / Time R	Delay time of each channel
Feedback	Number of repeats, up to 120 %. Above 100 % the loop self-oscillates; a soft limiter keeps it under control
Cross	Feeds each channel's repeats into the other
Width / Pan	Stereo width and balance of the echo
Drive	Saturation in the loop: Tape, Tube or Clip

Dial	What it does
Diffusion + Size	Smear the repeats into a cloud
Pitch	Shift per repeat, ±12 semitones (shimmer, octave down)
Ducking + Rel	Lowers the echo while the input is playing

Filters page

Four filters inside the loop: Low/High Cut, Bell, Low/High Shelf, Band Pass and Notch. Each repeat passes through them again, so the echo gets darker or thinner over time. The solid curve shows one pass; the dashed curve shows what is left after four repeats. Drag nodes to move them, scroll to change Q, double-click to add or toggle a filter, right-click to change its type. The analyser behind shows the input (dim) and the echo (bright).



Modulation page

Two LFOs (Sine, Triangle, Saw, Square, S&H, Smooth Random; rate in Hz or synced to tempo) and an Envelope follower drive up to six slots of **source → target → amount**. Targets include time, feedback, cross, filter frequencies, drive, diffusion, pitch, pan, width and mix.



Top bar and right column

Freeze holds the loop forever and stops recording new input. The right column has level meters and **Input**, **Mix** and **Output**. Up to 50 % Mix the dry signal stays at full level, so you can put Avel Time directly on a track. 20 factory presets; user presets go to `Documents/Avel/Avel Time/Presets`.

Avel Verb — algorithmic reverb

A high-quality algorithmic reverb with six algorithms, three era voicings and exact decay control in three frequency bands. Pick an algorithm in the segment at the top: **Room, Chamber, Hall, Plate, Ambience, Space**.



Reading the display

- The solid curve is decay time over frequency (left axis, seconds). Node **1** = Bass Decay, **2** = Decay, **3** = High Decay.
- The dashed green curve is the post EQ (right axis, dB): node **4** = Low Cut, **5** = High Cut.
- A live analyser runs behind the curves. The **Envelope** preview (top right) shows predelay, early reflections and the tail.
- Drag a node to edit it, Shift for fine moves, double-click to reset.



Controls

Control	Range	What it does
Color	1970s / 1980s / Now	Era voicing: 1970s is dark and chorused with soft saturation, 1980s is gritty and lush, Now is clean and full-bandwidth
Predelay	0–500 ms	Gap before the reverb starts
Size	0–100 %	Room size; early reflections scale with it
Decay	0.1–20 s	Reverb time (RT60) in the mid band
Diffusion	0–100 %	Density of the reflections
Early/Late	0–100 %	Balance of early reflections and tail
Mod Rate / Mod Depth	0.05–8 Hz / 0–100 %	Movement in the tail; removes metallic ringing
Width	0–150 %	Stereo width of the reverb
Ducking	0–100 %	Lowers the reverb while the input is playing
Bass / High Decay	×0.25–×4 / ×0.1–×2	Decay of lows and highs relative to Decay (nodes 1 and 3)

Control	Range	What it does
Low / High Cut	10 Hz–1 kHz / 1–20 kHz	Post EQ on the reverb, 12 dB/oct (nodes 4 and 5)

The right column has meters, **Input**, **Mix** and **Output**. **Freeze** (top right) holds the tail forever. Switching algorithms fades out and back in, so there are no clicks. 16 factory presets; user presets go to `Documents/Avel/Verb/Presets`.

Avel EQ — parametric equalizer

A parametric EQ for mixing and mastering: up to 24 bands, dynamic EQ, per-band Mid/Side, three processing modes, and tools that find problem frequencies and match a reference for you. Click a band to open its panel at the bottom.



Working with bands

Action	How
Add a band	Double-click the graph, or click the white curve (a "+" appears)
Move a band	Drag its point: left/right = frequency, up/down = gain

Action	How
Change Q	Mouse wheel over the point, or drag with Cmd (Ctrl on Windows)
Enable / disable a band	Double-click the point
Delete a band	Alt/Option-click the point, or press Delete
Select several	Shift-click or drag a box; then move them together
Band menu	Right-click the point (shape, slope, stereo, dynamics)

On Low/High Cut filters, vertical dragging changes resonance (Q).

Band panel

- **Shapes:** Bell, Low/High Shelf, Low/High Cut (6–96 dB/oct), Notch, Band Pass, Tilt Shelf.
- **Frequency, Gain, Q** dials.
- **Stereo / L / R / M / S** — process both channels, one side, Mid or Side only.
- Headphones = listen to this band alone; power = band on/off; bin = delete.
- **Dynamic** — makes the band react to the signal. **Threshold** sets where it starts, **Range** how far it moves.



Smart tools

- **Spectrum Grab** — hold the cursor over the spectrum: it freezes and highlights resonances with frequency, note and Q. Click a peak to create a matching band, then drag it down.
- **Sketch** — draw the curve you want with the mouse; Avel EQ places the bands for you.
- **Match EQ** — matches your track's tonal balance to a reference. **Learn** the reference by playing it through the plugin, or **Load File** (or drop an audio file onto the window). Set **Amount**, number of **Bands** and **Smoothing**, preview the curve, then **Apply**.



Top bar

- **Zero Latency / Natural / Linear** — processing mode. Zero Latency has analog-accurate curves with no delay; Natural adds analog-style phase (about 20 ms latency); Linear is linear phase.
- **Analyzer** — pre/post spectrum with a 4.5 dB/oct tilt; resolution, speed and Peak Hold are adjustable.
- **Auto Gain** keeps loudness steady while you boost or cut. **Output** sets the final level.

Avel Dynamic — compressor

A compressor with eight styles, a scrolling level history, a live transfer curve, sidechain filters and external sidechain, Mid/Side, lookahead, oversampling up to 4× and saturation. Latency is zero unless Lookahead or oversampling is on.



Reading the display

- Blue-grey fill is the input, white is the output. The red line from the top is gain reduction in dB.
- The teal line is the **threshold** — drag it, double-click to reset, scroll to step. The highlighted band around it is the knee.
- The **Transfer** window on the left shows the compression curve. The dot is the detector level and turns red while compressing. Drag up or down in it to change the threshold; scroll to change Ratio.

Controls

Dial	What it does
Threshold	Level above which compression starts (−60…0 dB)
Ratio	Amount of compression, 1:1 to ∞:1

Dial	What it does
Knee	Softness of the transition at the threshold (0…36 dB)
Range	Maximum gain reduction; Full = no limit
Attack / Release	Reaction speed (0.005…250 ms) and recovery (5 ms…2.5 s)
Hold	Holds the reduction before release starts
Lookahead	Lets the compressor see peaks early (up to 20 ms; adds latency)
Drive	Saturation amount when Character is Warm or Bright
Link	Channel link: 100 % = shared, 0 % = independent

Bottom row: **Auto Release** (fast after short peaks, slow on sustained compression), **Character** (Warm = even harmonics, Bright = odd harmonics up top), **Sidechain**, **Stereo / M/S / Mid / Side**, and **OS** (1×/2×/4×). The right column has IN / GR / OUT meters and **Input**, **Mix** and **Output**. Mix below 100 % gives parallel compression with no phase problems.

Styles

Style	Character
Clean	Transparent peak compressor; does exactly what you set
Classic	Feedback design with a soft knee, like classic VCAs
Opto	Slow, smooth optical-style levelling, programme-dependent release
Vocal	Wide automatic knee and RMS detector for steady vocals
Mastering	Gentle RMS, rounded attack, adaptive release
Bus	Glue for groups and the mix bus
Punch	Rounded attack lets transients through — drums snap
Pumping	Linear-scale recovery for audible pumping

Sidechain

Sidechain opens a panel with detector filters (High-pass, Low-pass), the **External** switch and **Listen**. To duck a bass from the kick, route the kick to the plugin's sidechain input in your DAW (in REAPER: the track's I/O button → send to channels 3/4), then switch on **External**.



Auto Gain keeps loudness equal while you compress, so you compare sound, not level. 22 factory presets in General, Vocals, Drums, Instruments and Mix Bus; user presets go to `Documents/Avel/Avel Dynamic/Presets`.

Avel Ess — de-esser

A de-esser that turns down harsh "s" and "sh" sounds without dulling the voice. In Split Band mode nothing outside the detection band changes, and the sound is untouched whenever there is no reduction.



Detection mode (top centre)

- **Single Vocal** compares the energy in the detection band with the whole signal, so bright vowels and breaths don't trigger it — only real sibilance does. Use it on a solo vocal.
- **Allround** reacts to the band level alone. Use it on mixes, drums, acoustic guitars, synths and reverb returns.

Reading the display

- **History** (top): input level (grey), detection level (teal, bright above the threshold) and gain reduction (red, hanging from the top) scroll by over about 3 seconds. Drag the threshold line to set it. **Max reduction** shows the peak value; click it to reset.
- **Band view** (bottom): the input spectrum with the detection band highlighted; it turns red while reducing. Drag the **L** or **H** edge, or drag the band itself to move both edges.



Controls

Control	What it does
Threshold	Level at which de-essing starts
Range	Maximum reduction, 0–24 dB
Release	How fast the reduction lets go, 5–250 ms
Lookahead	0–15 ms, so reduction is in place when the "s" starts
Link	Stereo link between channels
Band Low / Band High	Detection band: 1–16 kHz low edge, 2–20 kHz high edge (**Open** = no upper limit)
Processing	**Wide Band** turns the whole signal down on each "s"; **Split Band** turns down only the band
Channels	Stereo, Mid only, or Side only
Audition	Listen to the band being detected — handy for finding the "s"

The right column has IN / GR / OUT meters, **Input** and **Output**. 16 factory presets; user presets go to `Documents/Avel/Ess/Presets`. The window can be resized and remembers its size.

Avel Ember — multiband saturator

A multiband saturator with up to 6 bands, 17 saturation styles and oversampling up to 8×. Each band gets its own style and settings, so you can warm the lows, crunch the mids and add air on top independently. With no drive, the bands sum perfectly flat and colour nothing.



Working with bands

Action	How
Split into bands	Double-click the spectrum (or right-click → Split Band Here)
Move a crossover	Drag the vertical divider
Remove a crossover	Double-click the divider (or right-click → Remove)
Band drive	Drag the band's point up/down, or scroll over the band; Shift for fine moves

Action	How
Bypass a band	Double-click its point
Style, stereo, solo/mute	Band panel at the bottom, or right-click the band

Styles

Family	Styles
Saturation	Clean, Warm, Gentle
Tube	Clean, Warm, Heavy
Tape	Tape, Hot Tape
Transformer	Transformer
Amp	Clean, Crunch, Heavy
Clipping	Soft, Hard
FX	Rectify, Foldback, Bitcrush

Band panel

Dial	What it does
Drive	Saturation amount, 0–36 dB
Dynamics	Positive evens out the saturation; negative saturates loud parts harder
Tone	Brighter or darker character of the distortion
Feedback	Resonance around the band's centre
Mix / Level	Wet/dry blend and output level of the band

Also per band: **Stereo / M / S / L / R** processing, **S**olo, **M**ute and bypass. The **Transfer** window shows the live shape of the chosen style with a dot for the current signal level.



Top bar and right column

Oversampling Off / 2× / 4× / 8× reduces aliasing. **Auto Gain** keeps each band's loudness steady, so Drive changes the character, not the level. The right column has meters, **Input**, **Mix** (latency-aligned parallel blend, no comb filtering) and **Output**. 13 factory presets plus your own.

Avel Apex — mastering limiter

A mastering limiter with eight limiting styles, lookahead, two-stage release, oversampling up to 16×, true-peak limiting, EBU R128 loudness metering and dither. Put it **last** on the master channel.



Quick start

1. Set **Output** (bottom right) — the ceiling the signal will never exceed. For streaming use −1 dBTP with **True Peak** on.
2. Raise **Gain** (top right, or drag up/down directly on the display) until **Integrated** reaches your target. Pick the target in the **Target** pill; Integrated turns green within ±1 LU of it.
3. Choose a **Style** for the material. Check it with **Unity Gain** (hear the limiter at the original level, without the Gain boost) and **Audition** (hear only what the limiter removes).

Reading the display

- White fill and line = output level; blue-grey = input after Gain. The difference is what was limited.
- Red from the top = gain reduction.
- Yellow line = short-term loudness, yellow dashes = target, white dashes = ceiling.
- Top row: Short-term, Integrated, Range (LRA), True Peak, Max GR. Click to reset. 4 s / 8 s / 16 s sets the time scale.

Styles

Style	Character
Transparent	Cleanest: long S-shaped attack, programme-dependent release
Punchy	Keeps the drum click: shorter attack, fast release, a little clipping
Dynamic	Transients released very fast, the body held by the slow stage
Aggressive	Loud and forward: fast release, soft clipping of peaks
Allround	All-purpose default
Bus	Gentle glue for groups and stems
Safe	Minimum distortion: slow, smooth release
Modern	Dense and loud but controlled: adaptive two-stage release

Controls

Control	What it does
Lookahead (0–5 ms)	How early the limiter sees a peak. More = cleaner, less = more attack; 0 ms acts as a clipper
Attack	Speed of the slow stage on dense material
Release	Recovery speed; each style scales its fast and slow stages differently
Channel Link: Transients / Release	100 % = both channels limited together; lower = louder and wider, but the image can wander
Oversampling (Off…16×)	Limits at a higher rate for less aliasing
True Peak	Catches inter-sample peaks
DC Filter	Removes DC offset below 4 Hz
Dither	24/20/18/16-bit TPDF, with or without noise shaping. Use only when exporting to that bit depth

Factory presets in Mastering, Delivery, Mix and Utility; user presets go to `Documents/Avel/Apex Presets`. Latency is about 7 ms at 48 kHz and is compensated by your DAW.

Troubleshooting

Problem	Fix
The plugin doesn't show up in the DAW	Restart the DAW and run a plugin rescan. On Windows, make sure the DAW scans `C:\Program Files\Common Files\VST3`
macOS won't open the installer	System Settings → Privacy & Security → **Open Anyway** (macOS 15+), or right-click → Open (macOS 14 and older)
Windows shows a blue SmartScreen window	**More info → Run anyway**
Logic Pro doesn't list the plugins	Logic uses the AU version. Open Plug-in Manager and click **Reset & Rescan Selection**
A plugin adds delay	Lookahead, Linear/Natural Phase and oversampling add latency. Your DAW compensates it automatically
I want my presets on another computer	Copy the folder from `Documents/Avel` to the same place on the other computer

About Avel

Avel plugins are free. If they help your music, you can support the project with a donation on the Avel website — entirely optional.